INCH-POUND

MIL-T-82840(OS)
29 December 1989
SUPERSEDING
WS 21053A
21 July 1986
(see 6.6)

MILITARY SPECIFICATION

TAPE, SILICA CLOTH, PHENOLIC-PREIMPREGNATED

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 **Scope.** This specification defines one type of silica cloth tape preimpregnated with phenolic resin, referred to herein as the tape.

2. APPLICABLE DOCUMENTS

2.1 **Government documents.**

2.1.1 **Specifications, standards and handbooks.** The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Ordnance Station, Standardization Branch (3730), Indian Head, MD 20640-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 1340

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SPECIFICATIONS

MILITARY

MIL-R-9299	Resin, Phenolic, Laminating
MIL-B-82789	Broadgoods, Silica Cloth, Phenolic-Preimpregnated
MIL-S-82828	Silica Reinforcement
MIL-T-82846	Thread, Splicing, Nylon

STANDARDS

MILITARY

MIL-STD-129	Marking for Shipment and Storage

Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from: Military Specifications and Standards, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094).

2.1.2 **Other Government documents, drawings and publications.** The following other Government documents, drawings, and publications form a part of this document to the extend specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

CODE OF FEDERAL REGULATIONS (CFR)

49 CFR 100-199	Transportation

(Application for copies of CFRs should be addressed to the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402-0001).

2.2 **Non-Government publications.** The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2.).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM C 177	Steady-State Heat Flux Measurements and Thermal Transmission Properties by Means of the Guarded Hot Plate Apparatus (DoD adopted)
ASTM D 695	Compressive Properties of Rigid Plastics

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ASTM D 740 Methyl Ethyl Ketone (DoD adopted)

ASTM D 790 Flexural Properties of Unreinforced and Reinforced Plastics and Electrical Insulating Materials (DoD adopted)

ASTM E 228 Linear Thermal Expansion of Solid Materials with a Vitreous Silica Dilatometer

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187).

AMERICAN TRUCKING ASSOCIATION, INC.

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Association, Inc., Attn: Traffic Dept., 2200 Mill Road, Alexandria, VA 22314-4677.)

NATIONAL RAILROAD FREIGHT COMMITTEE

Uniform Freight Classification (UFC) 6000

(Application for copies should be addressed to the National Railroad Freight Committee, 222 South Riverside Plaza, Suite 1120, Chicago, IL 60606-5945.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services).

2.3 **Order of precedence.** In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 **First article.** When specified (see 6.2), a sample of the tape shall be subjected to first article inspection (see 6.3) in accordance with 4.3.

3.2 **Material.** The material shall be in the form of tape rolls cut from a woven silica cloth impregnated with phenolic resin in accordance with MIL-B-82789. The tape rolls shall be cut from preimpregnated broadgood rolls (see 6.4.1) on the bias, or an angle of 45 $\pm$ 5 degrees from the warp axis, and butt-spliced by sewing with nylon thread conforming to MIL-T-82846.

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3.2.1 **Resin.** The resin system shall conform to the requirements of MIL-R-9299, Grade A.

3.2.2 **Reinforcement.** The reinforcement shall be a fabric conforming to MIL-S-82828.

3.2.3 **Resin filler.** The resin filler shall be silica powder.

3.3 **Chemical and physical properties.** Chemical and physical properties of the tape shall conform to the requirements specified in tables I and II for uncured and cured material.

3.3.1 **Uncured material.** The physical properties of the uncured tape shall be in accordance with table I.

TABLE I. Uncured material chemical and physical properties.

Property	Percent by weight	
	Minimum	Maximum
Cloth content	52.0	65.0
Resin solids	29.0	36.0
Volatile content	3.0	5.5
Filler content	4.0	11.0
Resin flow	10.0	20.0

3.3.2 **Cured material.** The properties of the cured tape shall be in accordance with table II.

3.4 **Tape dimensions.** Tape width tolerance shall be ± 1/8 inch. The tape shall be wound on cores with an inside diameter of 3.0 inches. The outside diameter of the tape rolls shall not exceed 20 inches.

3.5 **Age of material.** The tape shall be shipped within 3 months after the manufacturing date of the broadgoods and shall be tested and certified within 60 days before shipment. The contractor shall certify that the material has been stored at a maximum temperature of 50°F between date of manufacture of the broadgoods and date of shipment. Temperature excursions above 50°F are permissible provided the cumulative time does not exceed 30 days.

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TABLE II. Cured material chemical and physical properties.

Property	Unit	Limits	
		Minimum	Maximum
Density	Grams/cubic centimeter (g/cc)	1.65	1.80
Residual volatiles	Percent	0.0	2.5
Resin content	Percent	27.0	37.0
Compressive strength	Pounds per square inch (psi)		
Warp direction		20,000	75,000
Fill direction (edgewise)		15,000	70,000
Interlaminar double shear strength	Psi	4,000	15,000
Thermal conductivity at 250 degrees Fahrenheit (°F), fill direction	British thermal unit (Btu)/(hour)(foot)(°F)		
With ply		0.2	0.60
Across ply		0.2	0.50
Coefficient of thermal expansion at 75 to 400°F	In./in.-°F x 10^{-6}		
Across ply		5.0	20.0
With ply		2.0	6.0
Flexural strength	Psi		
Warp direction		16,000	45,000
Fill direction		11,000	40,000

3.6 Shelf life. The tape specified herein shall have a shelf life of 9 months after date of manufacture of broadgoods material when stored at 50°F maximum in sealed watertight plastic bags. The cumulative time out of storage of the tape shall not exceed 30 days.

3.6.1 Shelf life extension. The shelf life of all tape rolls within a single storage container may be extended for one additional 6-month period

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provided the average value of three specimens from the beginning of one tape roll with the longest out of storage time from the storage container conforms to the following when tested as specified in 4.6.2.1 and 4.6.2.3.

	Minimum	Maximum
Volatile content, percent by weight	2.0	6.0
Resin flow, percent by weight	8.0	20.0

3.7 **Workmanship.** The tape shall have no holes, tears, or nonuniform weave and shall be free from dirt, grease, oil, and other contaminants.

4. QUALITY ASSURANCE PROVISIONS

4.1 **Responsibility for inspection.** Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 **Responsibility for compliance.** All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, not does it commit the Government to accept defective material.

4.2 **Classification of inspections.** The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

4.3 **First article inspection.** First article inspection shall consist of the tests specified in table III and the quality conformance inspections (see 4.4). Failure to meet all of the requirements specified herein shall be cause for rejection of the first article sample.

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TABLE III. First article inspection.

Tests	Requirement	Test Method
Cured Material		
Residual volatile content test	3.3.2	4.6.3.1.2
Compressive strength (fill) test	3.3.2	4.6.3.1.4
Thermal conductivity test	3.3.2	4.6.3.1.6
Coefficient of thermal expansion test	3.3.2	4.6.3.1.7
Flexural strength test	3.3.2	4.6.3.1.8

4.4 **Quality conformance inspection.** Quality conformance inspection shall consist of the examination and tests specified in table IV for cured and uncured material.

4.4.1 **Sampling for inspection.**

4.4.1.1 **Lot.** For inspection purposes, a lot shall consist of all tape rolls which are impregnated in one continuous production run, using one batch of the impregnator resin and additives mixed in one vessel at one time, and offered for acceptance at one time.

4.4.1.2 **Selection of sample rolls for uncured properties.** Each roll in the lot shall be tested for uncured properties at the beginning and at the end of the roll. Three specimens from each location shall be tested.

4.4.1.3 **Selection of sample rolls for cured properties.** Sample rolls of the tape shall be selected at random from each lot in accordance with table V for the inspections specified in 4.6.3.1.1 through 4.6.3.1.8. Sample material shall be selected as specified in 4.4.1.4. The test specimens shall be removed from the beginning of each sample roll.

4.4.1.4 **Sample material.** Sufficient material shall be selected at random from each sample container to perform all of the inspections specified in 4.6.1 through 4.6.3.1.8.

4.4.1.5 **Noncompliance of samples.** Failure of one or more of the sample rolls of material to meet all of the specified requirements shall be cause for rejection of the lot.

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TABLE IV. Quality conformance inspection.

Examination and tests	Requirement	Test Method
Uncured Material		
Visual examination	3.2, 3.4, 3.7 and section 5	4.6.1
Volatile content test	3.3.1	4.6.2.1
Filler content test	3.3.1	4.6.2.2
Resin solids, dry weight test	3.3.1	4.6.2.2
Cloth content test	3.3.1	4.6.2.2
Resin flow test	3.3.1	4.6.2.3
Cured Material		
Density test	3.3.2	4.6.3.1.1
Resin content test	3.3.2	4.6.3.1.3
Compressive strength (warp) test	3.3.2	4.6.3.1.4
Interlaminar shear test	3.3.2	4.6.3.1.5

TABLE V. Sampling for cured properties.

Number of rolls in the lot	Number of sample rolls to be selected for testing
1 to 3	All
4 to 40	3

4.5 Test results. The average results of three tests at each location for each property shall conform to the specified property requirements in section 3. Individual specimen test results shall also be recorded.

4.6 Methods of inspection. The following inspection methods shall be used. Unless otherwise specified in the method, all weights, volumes, temperatures, and times shall be measured to the nearest specified unit or decimal.

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NOTE: Reagent grade chemicals shall be used for chemical reactions in the conduct of all inspections defined herein. Solvents and indicators may be commercial nonreagent grade materials.

4.6.1 **Visual examination.** Each sample roll shall be visually examined for conformance to 3.2, 3.4, 3.7 and section 5.

4.6.2 **Uncured materials tests.**

4.6.2.1 **Volatile content test.** Volatile content of the uncured material shall be determined in accordance with the following:

a. Cut a 16 ± 2-square-inch specimen with a minimum width of 1.5 inches for testing broadgood rolls, or full tape width for testing tape rolls. Weigh to the nearest 0.001 gram (g) (W1).

b. Place in a recirculating oven and heat to 325 ± 10°F for 10 ± 1 minutes. Specimens should be placed to get adequate air circulation.

c. Remove specimen, place in a desiccator and cool to room temperature.

d. Remove from the desiccator and weigh the specimen to the nearest 0.001 g (W2).

e. Calculate percent volatiles as follows:

$$\text{Percent volatiles} = \frac{W1 - W2}{W1} \times 100$$

Where: W1 = uncured weight of specimen, g

W2 = final weight of specimen, g.

f. Record volatile content to the nearest 0.1 percent.

4.6.2.2 **Resin solids, filler content, and cloth content test.**

a. Place a 1.5 ± 0.1-g specimen weighed to the nearest 0.001 g (specimen weight) in a 250-milliliter (ml) beaker.

b. Calculate devolatilized specimen weight (W) as follows:

$$W = \text{specimen wt} - \frac{(\text{percent volatiles})(\text{specimen wt})}{100}$$

Where: percent volatiles is the value determined from the volatiles content (see 4.6.2.1)

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c. Add 50 ml of N,N dimethyl formamide (DMF) and bring to a boil. Gently stir the boiling DMF and preimpregnated cloth for a minimum of 5 minutes. Remove beaker from heat source and cool for a minimum of 5 minutes while stirring gently.

d. Decant and transfer the liquid solution to a second 250-ml beaker.

e. Rinse the remaining fabric with DMF several times to ensure complete extraction of resin and filler from fabric and add the rinses to a second 250-ml beaker.

f. Final-rinse the fabric with acetone to remove all traces of DMF.

g. Add all rinses to the second 250-ml beaker.

h. Dry a filter and a filtering crucible at 350 $\pm$ 10°F for a minimum of 30 minutes. Weigh crucible and filter to the nearest 0.001 g (W2).

i. Dry the fabric in an oven for 15 $\pm$ 1 minutes at 325 $\pm$ 10°F, cool in a desiccator to room temperature and reweigh to the nearest 0.001 g (W1).

j. Using an aspirator or vacuum pump, filter the contents of the second 250-ml beaker through a filter and filtering crucible into a vacuum flask.

k. Rinse the crucible and filter several times with acetone to remove all resin.

l. Dry the crucible and contents at 350 $\pm$ 10°F for 20 $\pm$ 1 minutes, cool in a desiccator to room temperature, and reweigh to the nearest 0.001 g (W3).

m. Calculate the resin solids content, filler content, and cloth content as follows:

$$\text{Percent resin solids content} = \frac{W - (W1 + W4)}{W} \times 100$$

$$\text{Percent filler content} = \frac{W4}{W} \times 100$$

$$\text{Percent cloth content} = \frac{W1}{W} \times 100$$

Where: W = calculated devolatilized weight of preimpregnated specimen, g

W1 = weight of fabric after resin and filler removal, g

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W2 = crucible and filter weight, g

W3 = filler + crucible and filter weight, g

W4 = W3 - W2 = filler weight

n. Record the resin solids, filler content, and cloth content to the nearest 0.10 percent.

4.6.2.3 Resin flow test.

a. Using 4-inch by 4-inch ± 1/8-inch template, cut four squares across width of fabric. Cut all squares on bias to eliminate fiber loss in testing. Stack squares uniformly on each other to make specimen.

NOTE: When retesting tape which is less than 4.0 inches for shelf life extension (see 3.6.1), it shall be permissible to use multiple tape widths which combine to provide a 4.0 inch specimen. Alternate tape layers shall be rotated 90 degrees.

b. Weigh specimen of four plies to the nearest 0.01 g, and record as W1.

c. Place specimen between release film. Preheat press to 325 ± 10°F, position specimen in middle of press plate, and apply press load of 150 ± 10 pounds per square inch gage (psig) immediately. Press-load specimen for a minimum of 10 minutes at 325 ± 10°F.

d. Remove specimen from press and cool to ambient temperature.

e. Using a knife, scrape off resin flash to original size of specimen. Do not remove any reinforcement from original dimensions.

f. Reweigh specimen to nearest 0.01 g and record as W2.

g. Calculate percent resin flow as follows:

$$\text{Percent flow} = \frac{(W1 - W2)}{W1} \times 100$$

Where: W1 = original weight of the specimen, g

W2 = final weight of the specimen, g

h. Record resin flow to nearest 0.001 percent.

4.6.3 Cured material tests.

4.6.3.1 Preparation of test panels. Prepare test panels of sufficient size to provide the specimens for all required tests as follows:

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NOTE: The outer 1/2 inch of the test panel shall not be used for preparation of specimens.

a. Cut sample material as required to fabricate the panel as specified in the individual test procedure, and cure plies into a panel between separator film on smooth caul plates as follows:

(1) Orient the plies such that the warp fibers of each ply are aligned in the same direction.

(2) Place the stack of plies into a press preset at 325 ± 10°F.

(3) Increase the pressure slowly to 1000 ± 10 pounds per square inch (psi), allowing the resin to stage to minimize flash at the ply edges.

(4) Hold the pressure and temperature for 120 ± 15 minutes, or 1 hour per inch of thickness, whichever is greater.

(5) Decrease the temperature and pressure to ambient.

4.6.3.1.1 Density test.

a. Cut sufficient plies to produce a test panel 0.125 ± 0.025 by 4 inches (minimum) by 5 inches (minimum), mark the warp direction and prepare panel as specified in 4.6.3.1.

b. Cut specimen 1 inch by 1 inch by panel thickness at random from the test panel. Wipe the specimen clean using methyl ethyl ketone (MEK) conforming to ASTM D 740, and allow to dry in air 20 minutes minimum before testing.

c. Weigh and record the weight of the specimen in air to the nearest 0.1 milligram (mg) (W1).

d. Weigh the specimen in water at 73 ± 3°F, and record the weight to the nearest 0.1 mg using a wire and sinker if required (W2).

e. Calculate density using water as follows:

$$\text{Density (g/cc)} = \frac{0.9975\ W1}{(W1 + W3 - W2)}$$

Where: W1 = weight of specimen in air, g

W2 = weight of specimen, wire, and sinker suspended in water, g

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W3 = weight of wire and sinker suspended in water to the same depth as in the W2 determination to the nearest 0.1 mg, g

0.9975 = conversion factor (specific gravity to density)

f. Record the density to the nearest 0.01 g/cc.

4.6.3.1.2 Residual volatile content test.

a. Cut sufficient plies to produce test panel 0.250 ± 0.050 thick by 4.000 ± 0.125 by 4.000 ± 0.125 inches. Prepare panel as specified in 4.6.3.1 with the following exception:

Apply contact pressure for 30 ± 5 seconds and dump pressure for two cycles before slowly applying maximum pressure. (The material shall not be staged.)

b. Cut a specimen 1.000 ± 0.050 inch by 1.000 ± 0.050 inch by panel thickness from the center 2-inch by 2-inch section of the test panel. Top and bottom molded surfaces shall not be machined. Wipe the specimen clean using MEK conforming to ASTM D 740, and allow to air dry 20 minutes minimum before testing.

c. Place the specimen in a desiccator and desiccate for 18 hours minimum.

d. Weigh the specimen to nearest 0.001 g and record as W1.

e. Place the specimen into an air circulating oven which has been preheated and stabilized at 325 ± 10°F for approximately 30 minutes. Condition the specimen at 325 ± 10°F for 4 to 4.5 hours.

f. Remove the specimen from the oven and cool in a desiccator for approximately 30 minutes or until the specimen reaches room temperature.

g. Reweigh the specimen to the nearest 0.001 g. Record as W2.

h. Calculate the percent residual volatiles as follows:

$$\text{Percent volatiles} = \frac{(W1 - W2)}{W1} \times 100$$

Where: W1 = original weight of specimen in g

W2 = devolatilized weight of specimen in g

i. Record the residual volatiles to the nearest 0.001 percent.

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4.6.3.1.3 Resin content test. The resin content of the cured material shall be determined in accordance with the following:

a. Weigh a 5.5 ± 0.5-g specimen (see 4.6.3.1) to the nearest 0.001 g in a previously fired (1450°F) and tared crucible.

b. Devolatilize the sample by conditioning the sample and crucible in a circulating air oven at 325 ± 10°F for 20 ± 1 minutes, cool in a desiccator, and reweigh to the nearest 0.001 g to obtain the sample weight. Designate this weight as W1.

c. Place the sample and crucible into a muffle furnace.

d. Condition at 1450 ± 10°F for 3 hours minimum.

e. Cool in a desiccator to room temperature, and reweigh to the nearest 0.001 g to obtain final sample weight. Designate this weight as W2.

f. Calculate resin content as follows:

$$\text{Percent resin solids} = \frac{(W1 - W2)}{W1} \times 100$$

Where: W1 = sample weight after volatile removal, g

W2 = sample weight after volatile removal, and 3 hours minimum at 1450 ± 10°F, g.

g. Record resin content to the nearest 0.1 percent.

4.6.3.1.4 Compressive strength test. The compressive strength shall be determined, using the test panel prepared in 4.6.3.1.1, in accordance with the following:

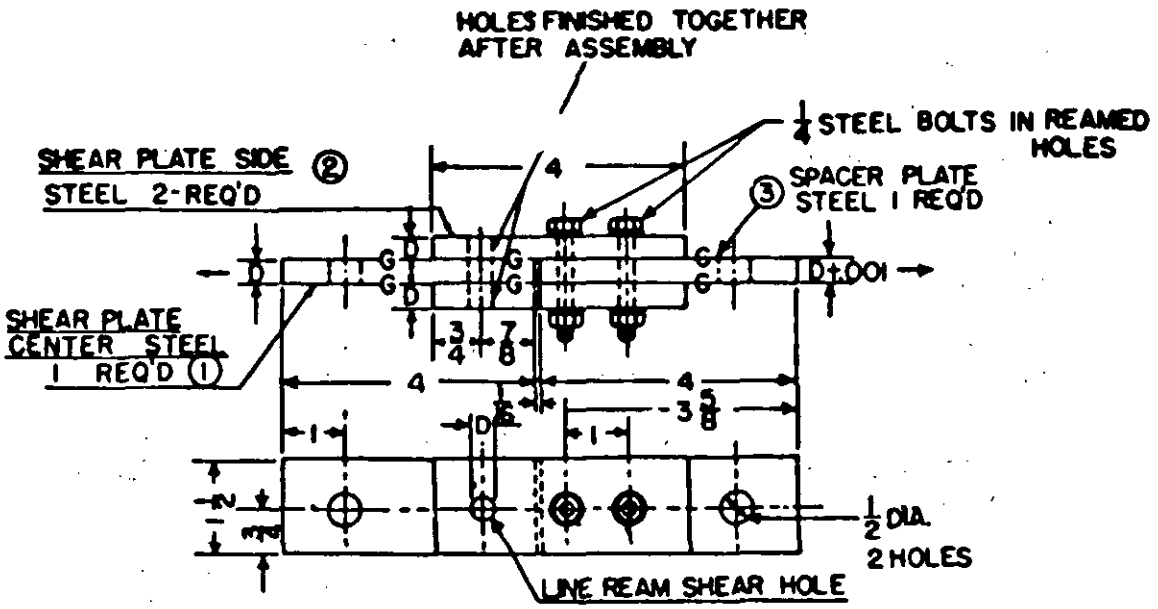
a. Machine specimen to dimensions shown on figure 1.

b. Measure width and thickness of specimen to nearest 0.001 inch in gauge section area. Calculate and record cross-section area.

c. Place specimen in support jig (see Figure 4 of ASTM D 695) centered so the ends project an even amount beyond both ends of the jig. Tighten finger tight only.

d. Place the specimen and support jig in a testing machine and align so the specimen ends are parallel with surface of the compression tool. Adjust the crosshead of the testing machine until it just contacts the top of the specimen.

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NOTES

1. D is nominal diameter of test cylinder.
2. Finish: machine after heat treatment.
3. Grind surfaces denoted G.
4. Spacer plate (3) to be 0.001 inch thicker than (1).
5. Edges of shear hole must be sharp.
6. All dimensions in inches; tolerances—fractions ± 1/64; specimen diameter shall be D + 0.000, — 0.001; diameter shear hole shall be D + 0.000, — 0.001.
7. Alinement of shear holes must be maintained.
8. Material: steel, heat-treated to Rockwell C 35–40.

FIGURE 1. Compressive strength test specimen.

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e. Set the testing machine crosshead speed to 0.050 ± 0.010 inch per minute, and start the machine.

f. Record the maximum load carried by the specimen during the test.

g. Calculate the compressive strength by dividing the maximum load by the original cross-sectional area.

h. Record the compressive strength to the nearest 10 psi.

4.6.3.1.5 Interlaminar shear test.

a. Cut sufficient plies to produce a test panel 1.50 ± 0.25 inch thick across plies by minimum dimensions of 1.5 inch by 1.5 inch.

b. Spray the mold with release agent and layup plies into the mold.

c. Cure as specified in 4.6.3.1.

d. After cure, remove the mold from the press, remove the test panel from the mold, and allow the panel to cool to room temperature.

e. Cut a specimen of sufficient size to produce 0.375-inch-diameter by 1.125-inch-length (minimum) finished specimen with the length being perpendicular to the plies from the test panel.

f. Center the specimen to get the length of the cylindrical specimen perpendicular to the plies, and machine the specimens to a diameter of 0.375 + 0.000/-0.002 inch.

g. Measure the diameter of the test specimen to 0.0001 inch and calculate the cross-sectional area (A).

h. Place the specimen in a tensile loading shear jig conforming to figure 2.

i. Place the shear fixture in the testing machine, align, and pin the ends of the fixture.

j. Set the machine at an indicated crosshead speed of 0.015 to 0.025 inch per minute, and test.

k. Record the maximum load carried by the specimen during the test, and record as L.

l. Calculations:

$$\text{Interlaminar shear (psi)} = \frac{L}{2A}$$

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Where: L = maximum load carried by specimen during testing, in pounds

A = original cross-sectional area of specimen, in square inches

m. Record the interlaminar shear to the nearest 10 psi.

4.6.3.1.6 **Thermal conductivity test.** Thermal conductivity shall be determined in accordance with ASTM C 177 and as follows:

a. Prepare a test panel of the required size, and cure as specified in 4.6.3.1.

b. Data shall be recorded for temperature of 250°F.

c. Maximum test panel size shall be 9.00 inches. Wrinkles are not permitted in the "with ply" specimens.

d. Record thermal conductivity to nearest 0.01 Btu/ft-hr-°F.

4.6.3.1.7 **Coefficient of thermal expansion test.** The coefficient of thermal expansion (see 6.4.4) shall be determined in accordance with ASTM E 228 and as follows:

a. Prepare test panel of required size, and cure as specified in 4.6.3.1.

b. Data shall be recorded for temperature range from 75 to 400°F.

c. Maximum test panel specimen length shall be 4.0 inches.

d. Record coefficient of thermal expansion to the nearest 0.01 in./in.-°F x 10^{-6}.

4.6.3.1.8 **Flexural strength test.** Flexural strength shall be determined in accordance with ASTM D 790, Method I, and as follows:

a. Prepare test panel of required size, and cure as specified in 4.6.3.1.

b. Maximum test specimen thickness shall be 0.25 inch.

c. Record flexural strength to the nearest 10 psi.

4.7 **Inspection of packaging.** The tape containers and container markings shall be visually examined to verify compliance with section 5.

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5. PACKAGING

5.1 Packaging. Unless otherwise specified in the contract or order (see 6.2), packaging shall be level C as specified herein.

5.1.1 Level C. The tape shall be packaged to afford adequate protection against loss, contamination, deterioration and damage during shipment from the supply source to the first receiving activity and during storage under the shelf life period and conditions specified in 3.6. Containers in the same shipment shall be of the same size. Each tape roll shall be separately wrapped in a plastic bag. The packaging shall conform to UFC 6000, National Motor Freight Classification, 49 CFR 171-178, or to other carrier rules and regulations as applicable to the mode of transportation.

5.2 Marking

5.2.1 Standard marking. In addition to any special marking required by the contract or order (see 6.2), the tape shall be marked in accordance with MIL-STD-129.

5.2.2 Special marking. In addition to the marking requirements of 5.2.1, the tape shall include the following information:

a. Title, number and date of this specification.

b. Manufacturer's name and address.

c. Material trade name and identification.

d. Number of rolls and net weight of contents.

e. Lot number and date of quality conformance inspection.

f. Shelf life and inspection/test date for extension. In addition, the marking shall include the following:

"SHELF LIFE EXTENSION

THE SHELF LIFE OF AN INDIVIDUAL LOT MAY BE EXTENDED FOR ONE ADDITIONAL 6-MONTH PERIOD PROVIDED THE VOLATILE CONTENT AND RESIN FLOW, UPON RETEST, CONFORMS TO THE REQUIREMENTS OF MIL-T-82840."

g. Mark opposite sides of each shipping container with the following:

"TEMPERATURE SENSITIVE MATERIAL.
DO NOT EXPOSE TO TEMPERATURE HIGHER THAN 100°F."

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5.3 Shipment temperature. The temperature of the silica cloth during shipment shall not exceed 100°F as measured by temperature-indicating tape placed on the inside of the spool of at least one tape roll in each container in each shipment.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The silica cloth tape specified herein is intended for use in the manufacture of Standard Missile Mark 104 Dual Thrust Rocket Motor nozzles.

6.2 Acquisition documents. Acquisition documents must specify the following:

a. Title, number, and date of this specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1, 2.1.2 and 2.2).

c. Whether first article is required (see 3.1 and 6.3).

d. Tape dimensions (see 3.4).

e. Special marking, if other than as specified in 5.2.2.

6.3 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors on the material to be inspected. The material should be a first article sample and should consist of a quantity of the tape sufficient to perform all of the inspections specified in 4.3 and 4.4. The first article sample should be packaged and marked as specified in section 5. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.4 Definitions.

6.4.1 Broadgoods roll. A wide cloth roll received from the cloth supplier and impregnated with resin.

6.4.2 Specimen. A portion of material selected from a sample roll for inspection.

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6.4.3 **Tape roll.** A narrow tape cut from a master roll of impregnated cloth.

6.4.4 **Coefficient of thermal expansion.** The secant slope between 75 and 400°F of the linear expansion.

6.4.5 **Sample.** A broadgoods roll of impregnated cloth selected in accordance with the requirements of this specification for performance of inspections.

6.4.6 **Stage.** A method of stabilizing and retaining the phenolic polymer in a solid state on the carbon cloth by temperature cycling.

6.4.7 **Bias.** The angular orientation of the warp and fill of the cloth weave relative to the longitudinal axis of the cut tape.

6.5 **Subject term (key word) listing.**

Rocket Motor, Dual Thrust, Mark 104
Standard Missile

6.6 **Supersedure information.** MIL-T-82840 incorporates the following engineering change proposal (ECP) and notice of revision:

ECP	NOR
MTA060 (11/10/88)	MTA060.3 (2/7/88)

Preparing activity:
NAVY-OS
(Project 1340-N812)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-T-82840	2. DOCUMENT DATE (YYMMDD) 29 December 1989

3. DOCUMENT TITLE
TAPE, SILICA CLOTH, PHENOLIC-PREIMPREGNATED

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY NAVAL ORDNANCE STATION (CODE 3730) INDIAN HEAD, MD 20640-5000

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON
c. ADDRESS (Include Zip Code)	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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